



CONTEMPORARY CLIMATE CHANGE

Observed and Projected Changes

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OBSERVED CHANGES

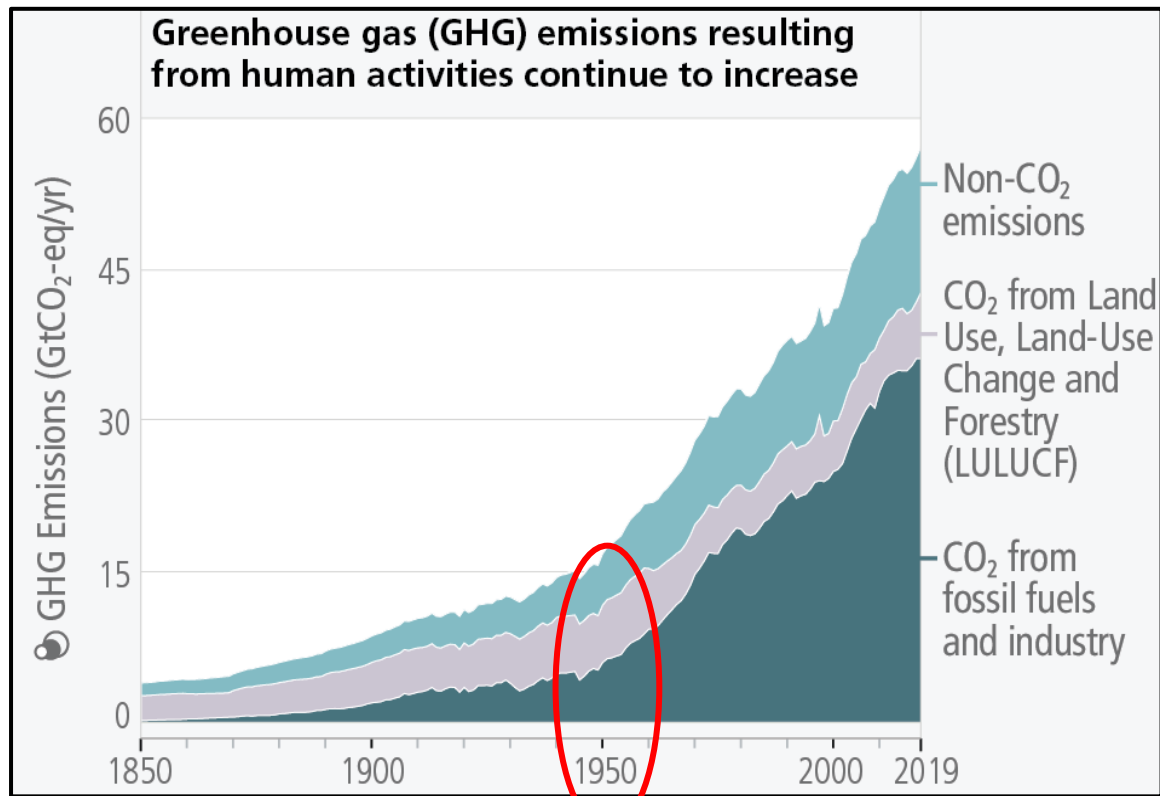
Where Are We Now?

OBSERVED CHANGES

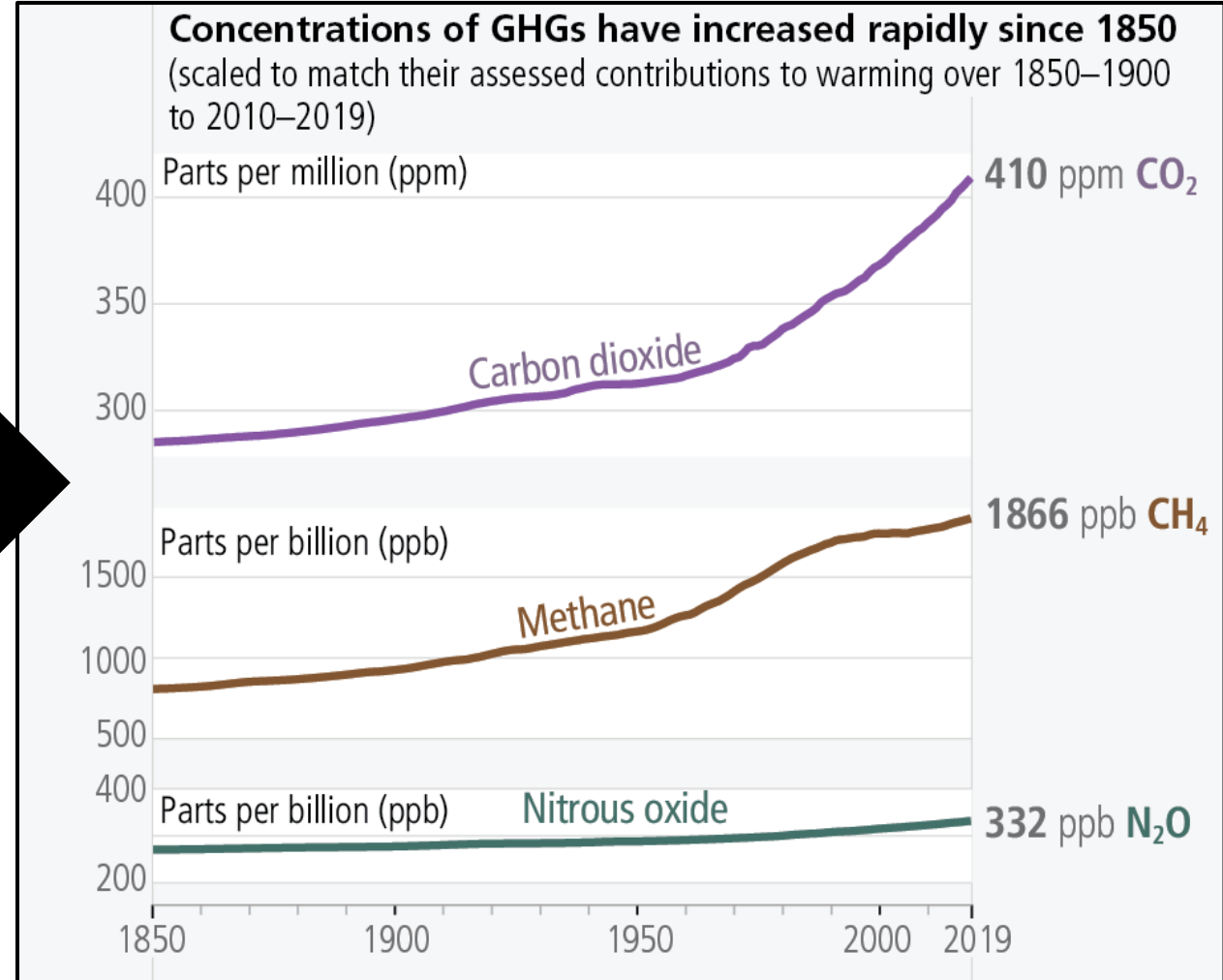
Greenhouse Gas Emissions

Increased concentrations of GHGs in the atmosphere

Increased emissions of GHGs



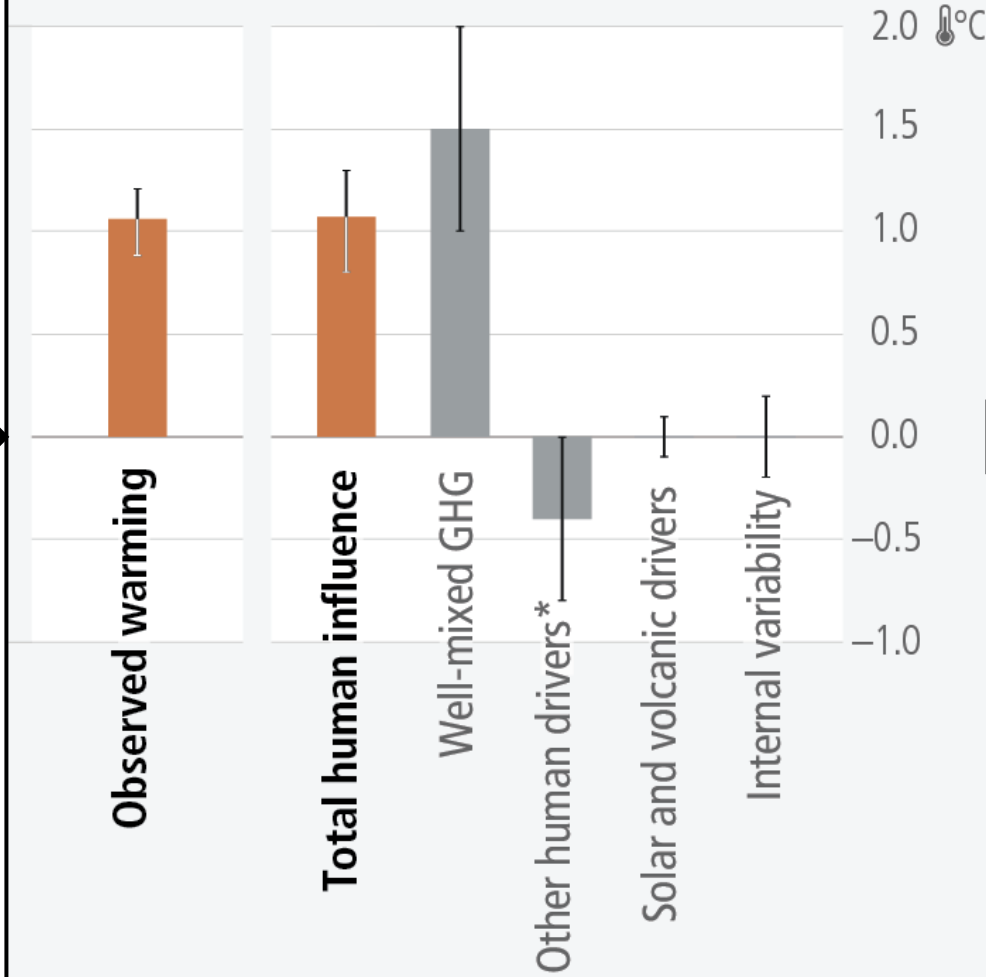
Great Acceleration ~ post world war II Industrialization etc.



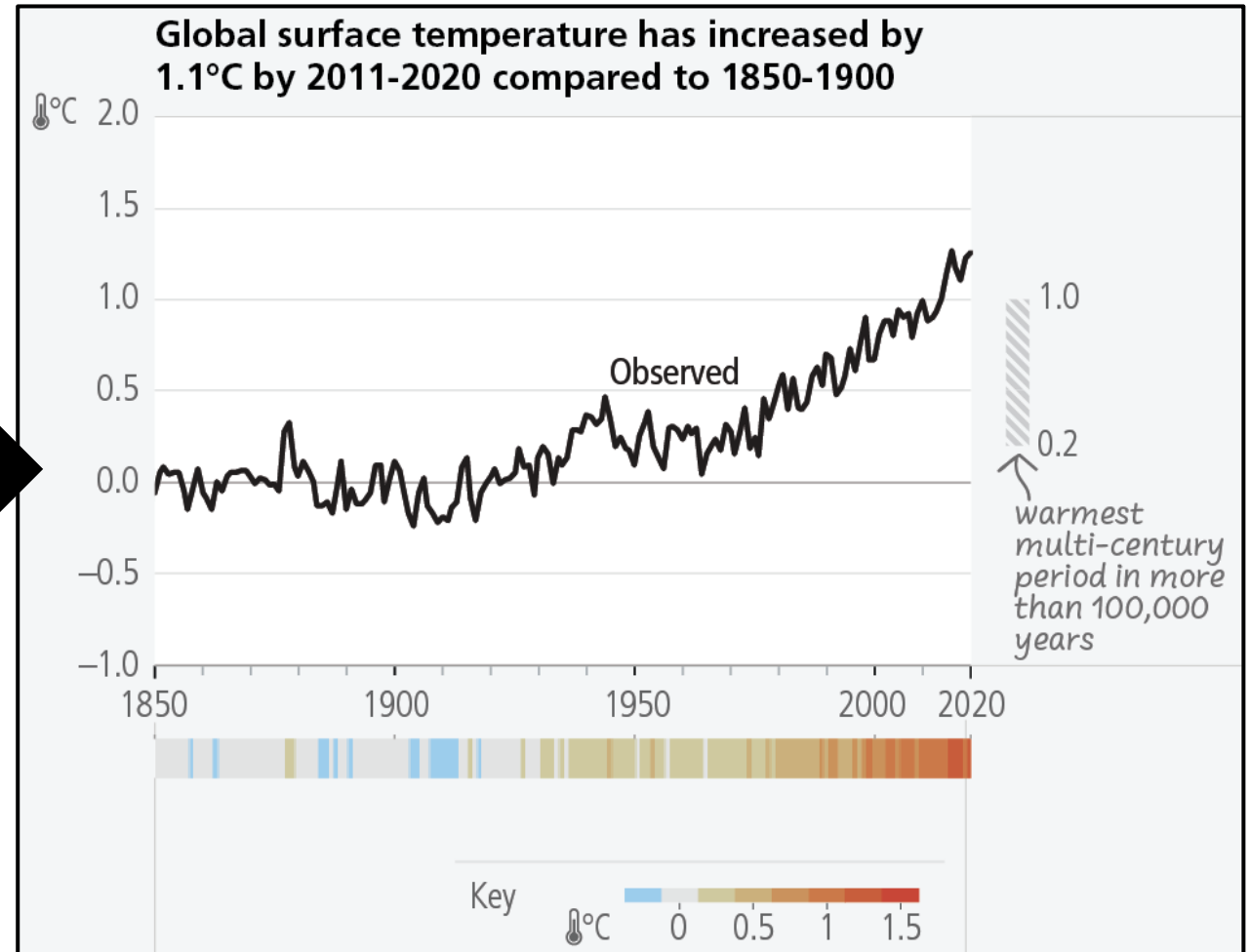
OBSERVED CHANGES

Temperature

Observed warming is driven by emissions from human activities with GHG warming partly masked by aerosol cooling 2010–2019 (change from 1850–1900)

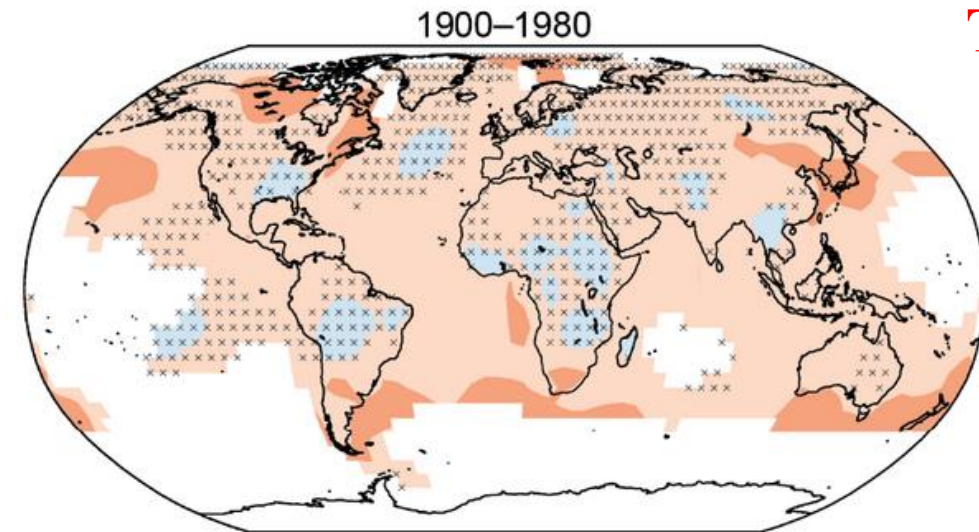


Changes in global surface temperature

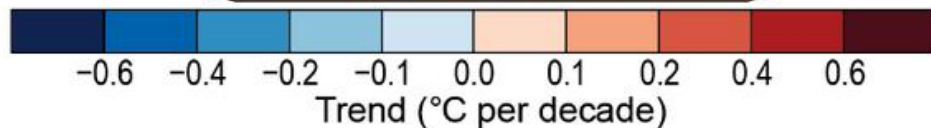
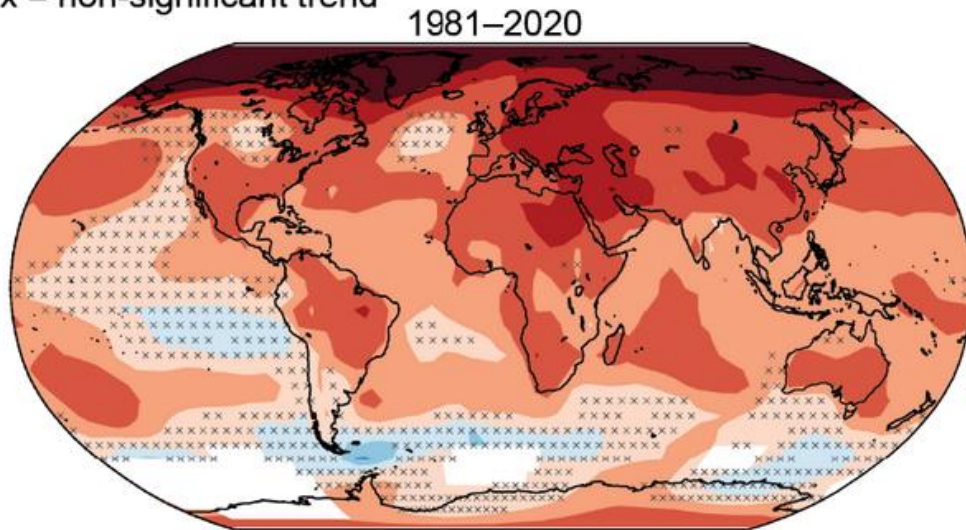


OBSERVED CHANGES

Temperature



x = non-significant trend



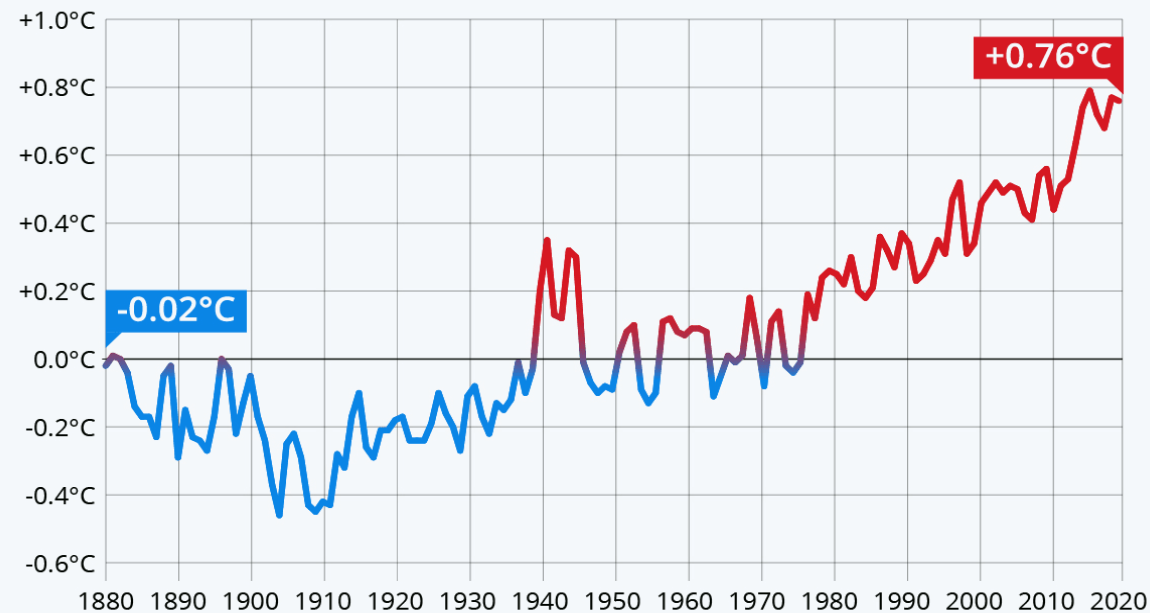
- Global surface temperature was 1.1°C higher in 2011–2020 than 1850–1900
- Has increased faster since 1970 than in any other 50-year period over at least the last 2000 years
- Each of the last four decades has been successively warmer than any decade that preceded it since 1850

OBSERVED CHANGES

Temperature

The Oceans Are Getting Warmer

Annual divergence of global ocean temperature
from 20th century average (1880-2020)



Ocean surface temperatures

Source: NOAA National Centers for Environmental Information (NCEI)

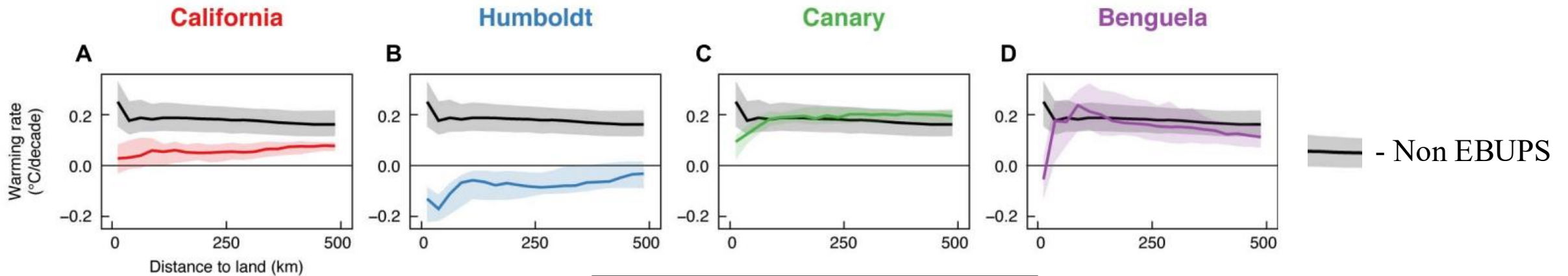


OBSERVED CHANGES

Temperature

Regional Nearshore Cooling

- Nearshore cooling associated with Eastern Boundary Upwelling Systems
- Wind-driven coastal upwelling brings deep cold, nutrient-rich water to the surface



Reduced Nearshore Warming Associated With Eastern Boundary Upwelling Systems

Rui Seabra¹, Rubén Varela², António M. Santos^{1,3}, Moncho Gómez-Gesteira²,
Cláudia Meneghesso^{1,3}, David S. Wetthey⁴ and Fernando P. Lima^{1*}

Most eastern boundary
upwelling regions represent
thermal refugia in the age of
climate change

Marisol García-Reyes^{1*}, Gammon Koval¹, William J. Sydeman¹,
Daniel Palacios², Luis Bedriñana-Romano³, Kelly DeForest⁴,
Carlos Montenegro Silva⁵, Maritza Sepúlveda⁶ and Ellen Hines⁴

Under Pressure: Climate Change, Upwelling, and Eastern Boundary Upwelling Ecosystems

Marisol García-Reyes^{1*}, William J. Sydeman¹, David S. Schoeman²,
Ryan R. Rykaczewski³, Bryan A. Black⁴, Albertus J. Smit⁵ and Steven J. Bograd⁶

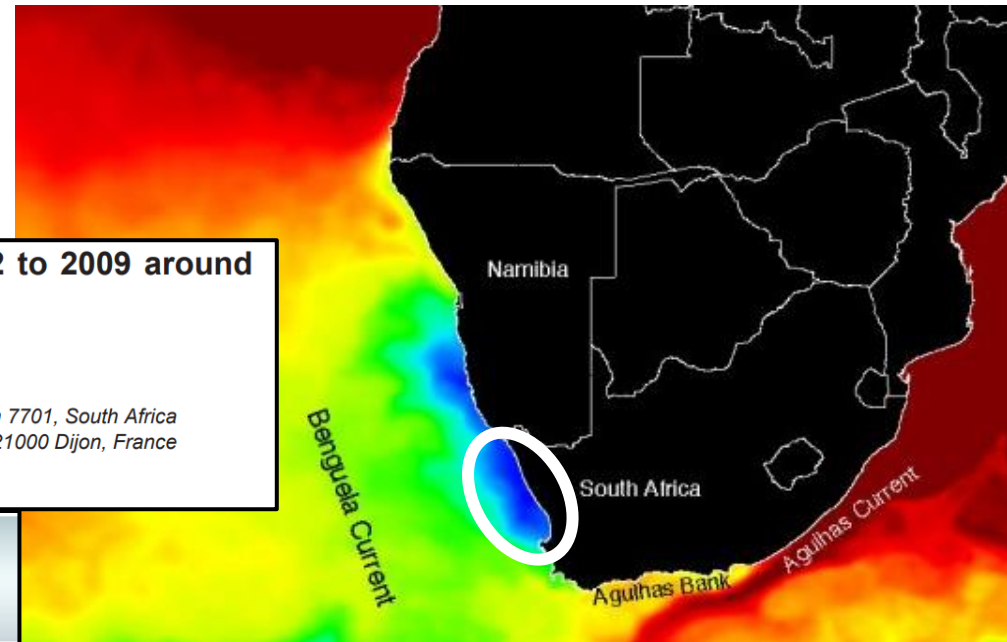
OBSERVED CHANGES

Temperature

Regional Nearshore Cooling

South African west coast:

- 0.5°C cooling per decade
- Largely driven by upwelling



Coastal oceanic climate change and variability from 1982 to 2009 around South Africa

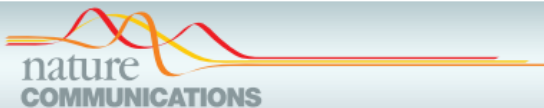
M Rouault^{1*}, B Pohl² and P Penven³

¹ Department of Oceanography, Marine Research Institute, University of Cape Town, Rondebosch 7701, South Africa

² Centre de Recherches de Climatologie, CNRS / Université de Bourgogne, 6 boulevard Gabriel, 21000 Dijon, France

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Three decades of high-resolution coastal sea surface temperatures reveal more than warming

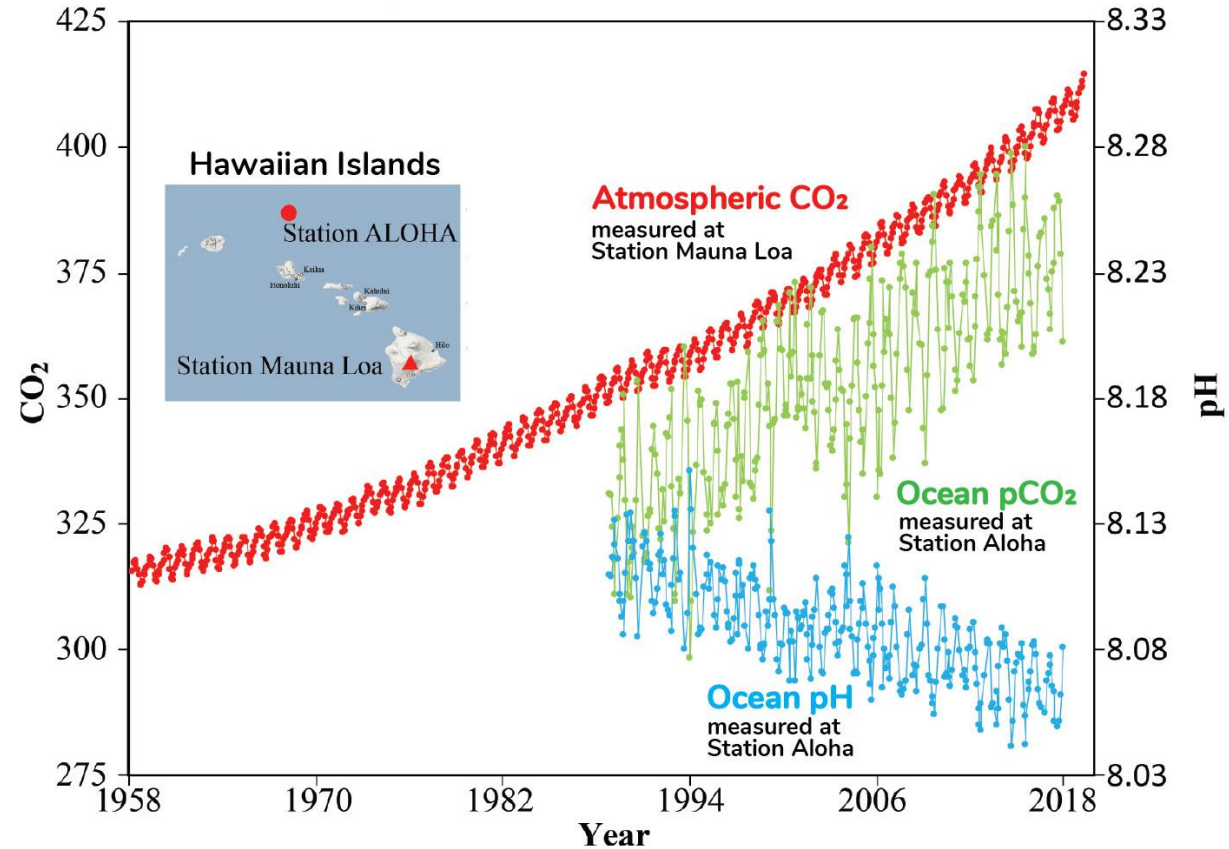
Fernando P. Lima^{1,2} & David S. Wetthey¹

Upwelling indices for comparative ecosystem studies: Variability in the Benguela Upwelling System

T. Lamont^{a,b,*}, M. García-Reyes^c, S.J. Bograd^d, C.D. van der Lingen^{e,f}, W.J. Sydeman^{c,g}

OBSERVED CHANGES

Ocean Acidification

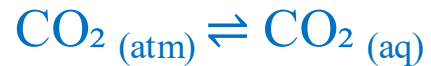


- The ocean acts as a major carbon sink, absorbing a significant portion of the excess atmospheric CO_2
- This causes a decrease in ocean pH and alters the overall global ocean chemistry ~ ocean acidification

OBSERVED CHANGES

Ocean Acidification

- **CO₂ Absorption:** The ocean is a carbon sink, absorbs a significant portion of the excess CO₂ released into the atmosphere
- **Carbonic Acid Formation:** CO₂ reacts with seawater to form carbonic acid (H₂CO₃)
- **Dissociation and pH Decrease:** H₂CO₃ dissociates into bicarbonate (HCO₃⁻) and hydrogen ions (H⁺)
- **Further dissociation:** Bicarbonate HCO₃⁻ further dissociates into H⁺ and carbonate (CO₃²⁻) ions
- **Acidification:** Increase in H⁺ concentration → decrease in pH
 - The availability of CO₃²⁻ declines
 - While carbon increases as more CO₂ dissolves in seawater, the concentration of CO₃²⁻ decreases



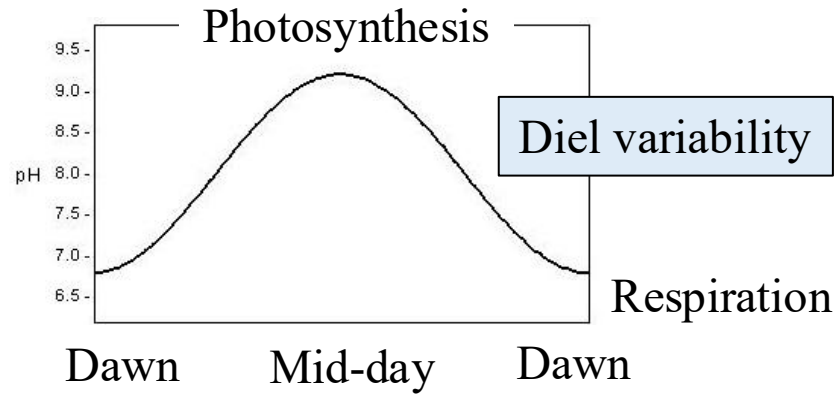
- Eqs. 2, 3 produce protons (H⁺)
- Lowers the pH
- $\text{pH} = -\log_{10}[\text{H}^+]$

OBSERVED CHANGES

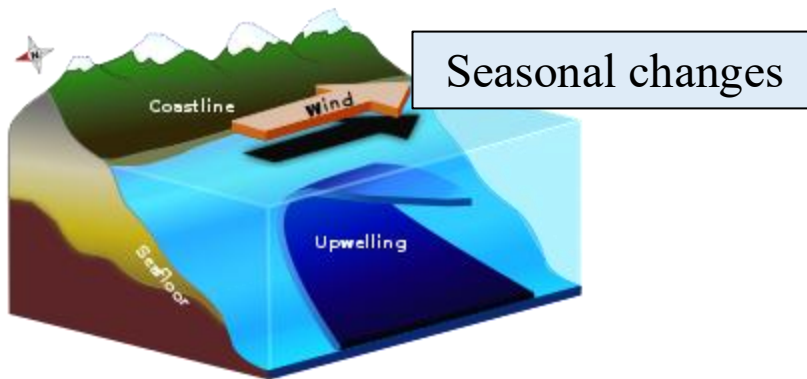
Ocean Acidification

Variability in carbonate chemistry in nearshore and coastal environments

Temporal variability



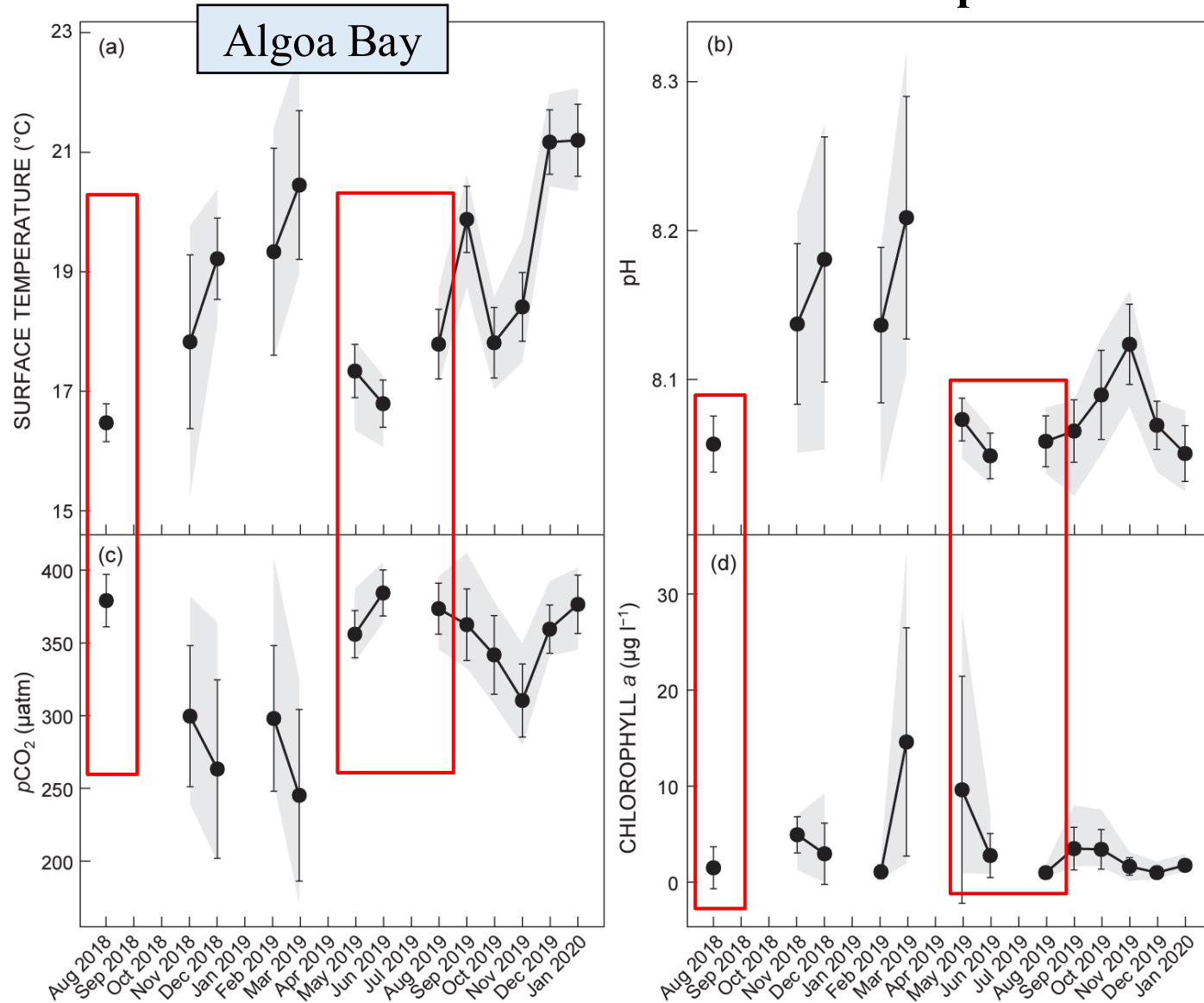
- Photosynthesis during the day utilises CO_2 , which increases pH
- Respiration at night replenishes CO_2 , which decreases pH
- Upwelling brings cold nutrient-rich water to the surface



OBSERVED CHANGES

Variability in carbonate chemistry in nearshore and coastal environments

Temporal variability



A baseline assessment of coastal pH variability in a temperate South African embayment: implications for biological ocean acidification research

C Edworthy^{1,2,3*}, WM Potts^{2,1}, S Dupont^{4,5}, MI Duncan^{2,6,7}, TG Bornman^{8,3} and NC James^{1,2}

¹ South African Institute for Aquatic Biodiversity (SAIAB), Makhanda, South Africa

² Department of Ichthyology and Fisheries Science, Rhodes University, Makhanda, South Africa

³ Institute for Coastal and Marine Research, Nelson Mandela University, Gqeberha, South Africa

⁴ Department of Biological and Environmental Sciences, University of Gothenburg, Fiskebäckskil, Sweden

⁵ Radioecology Laboratory, International Atomic Energy Agency (IAEA), Marine Laboratories, Principality of Monaco

⁶ Department of Environment, University of Seychelles, Mahe Island, Seychelles

⁷ James Michel Blue Economy Research Institute, University of Seychelles, Mahe Island, Seychelles

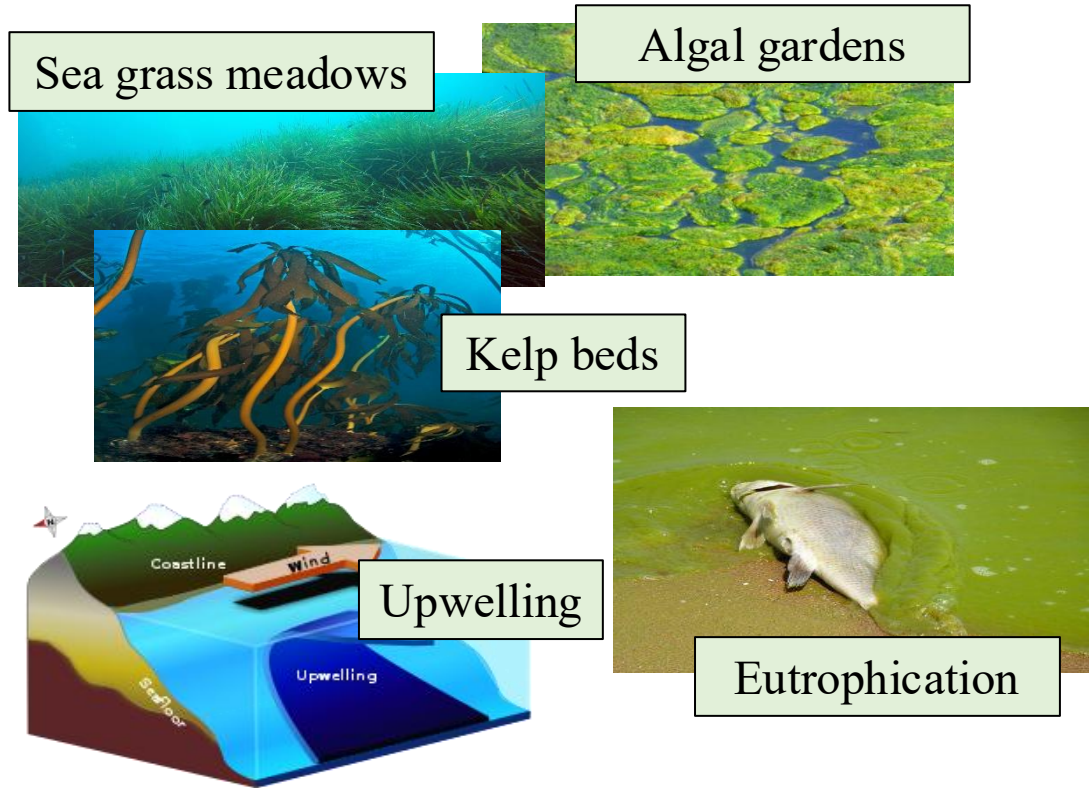
⁸ South African Environmental Observation Network, Elwandle Coastal Node, Gqeberha, South Africa

- $p\text{CO}_2$ is highest in winter when $T^\circ\text{C}$ is low
- Results in low seawater pH and reduced productivity during this period
- Lower $T^\circ\text{C}$ in winter reduces photosynthesis rates → decreased oxygen production and increased CO_2 absorption, lowering pH

OBSERVED CHANGES

Variability in carbonate chemistry in nearshore and coastal environments

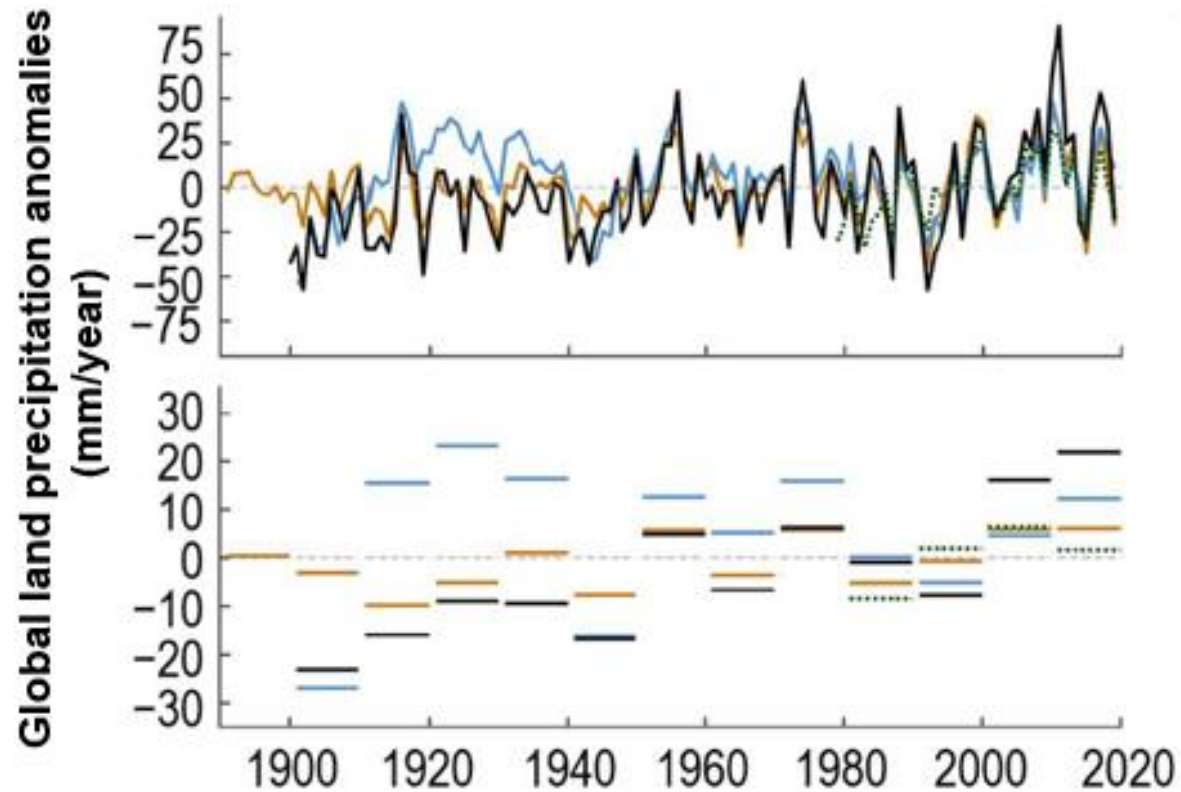
Spatial variability



- Spatially, variability stems from different sources
- In plant dominated ecosystems pH can vary from local to regional scales
- Driven by abiotic forcing such as freshwater input and upwelling
- Eutrophication (nutrient enrichment resulting in increased plant and algae growth) provides conditions for increased heterotrophic respiration, which replenishes CO₂

OBSERVED CHANGES

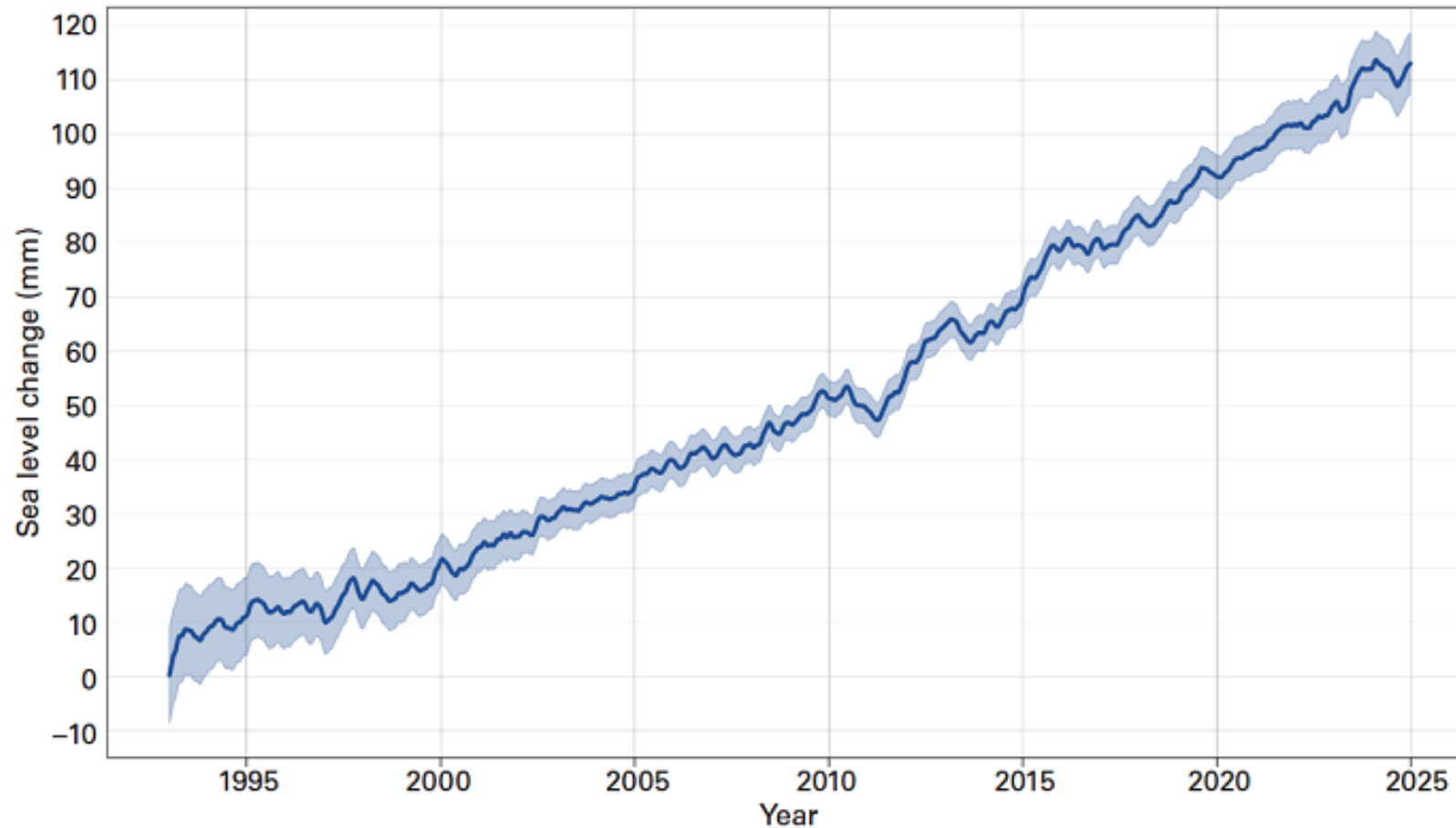
Precipitation



- Increase in air temperature leads to higher moisture-holding capacity of the atmosphere
- Enhances the hydrological cycle, altering the characteristics of precipitation amount, frequency, intensity, duration, type, etc
- Globally averaged precipitation over land has increased since 1950, with a faster rate of increase since the 1980s

OBSERVED CHANGES

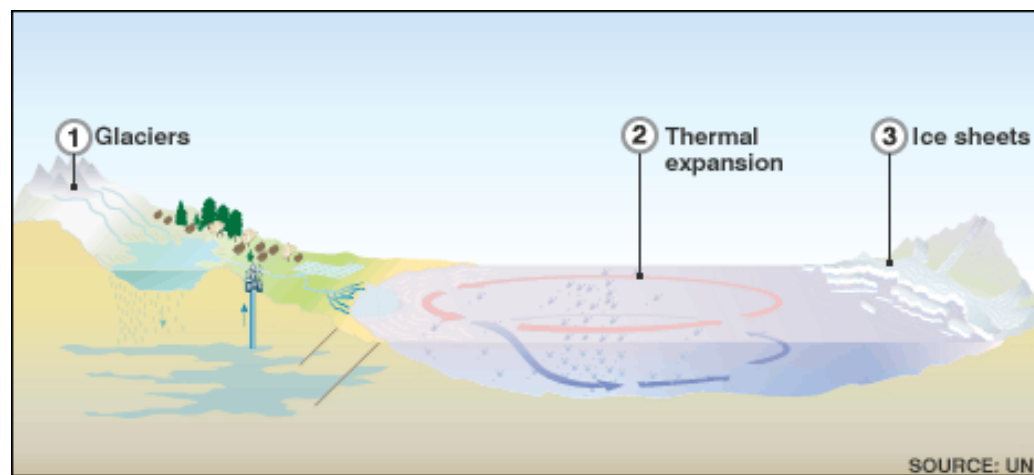
Rising Mean Sea Level



- The average rate of sea level rise has more than doubled, increasing from 2.1 mm yr^{-1} (1993-2002) to 4.7 mm yr^{-1} (2015-2024)
- In 2024, global mean sea level attained a new high in the satellite record (from 1993 to present)

OBSERVED CHANGES

Rising Mean Sea Level



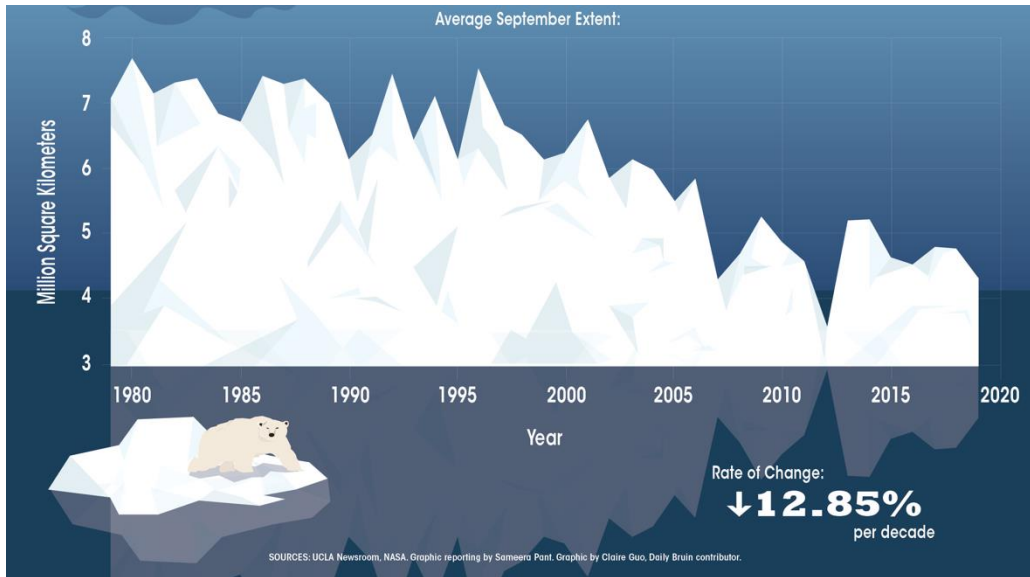
Thermal Expansion

- When water warms, its molecules move faster and take up more space, causing the water to expand
- This expansion, even without an increase in the water's mass, results in the water taking up more volume



OBSERVED CHANGES

Rising Mean Sea Level



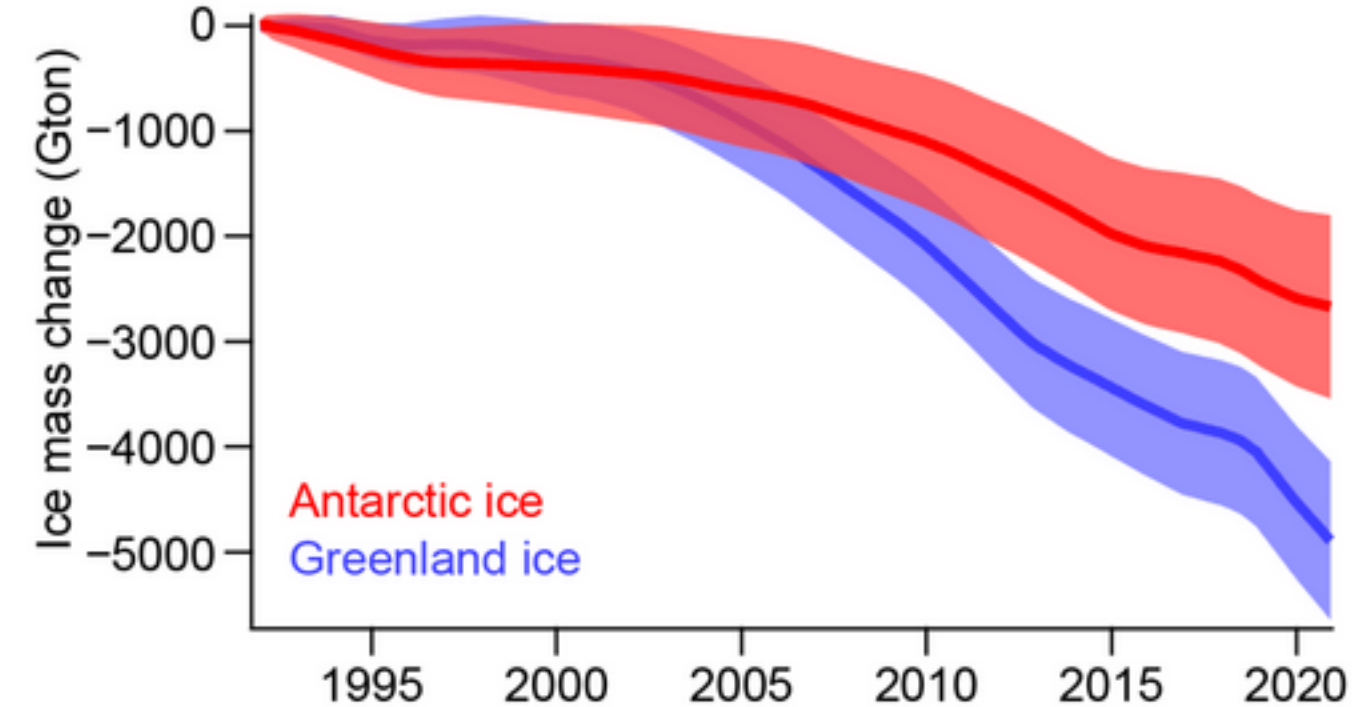
Melting Ice Caps

- Warmer air and ocean temperatures cause glaciers and ice sheets to melt faster, leading to a decrease in their volume
- Melting glaciers and ice sheets add freshwater to the oceans, further increasing their volume and sea level



OBSERVED CHANGES

Ice and Snow

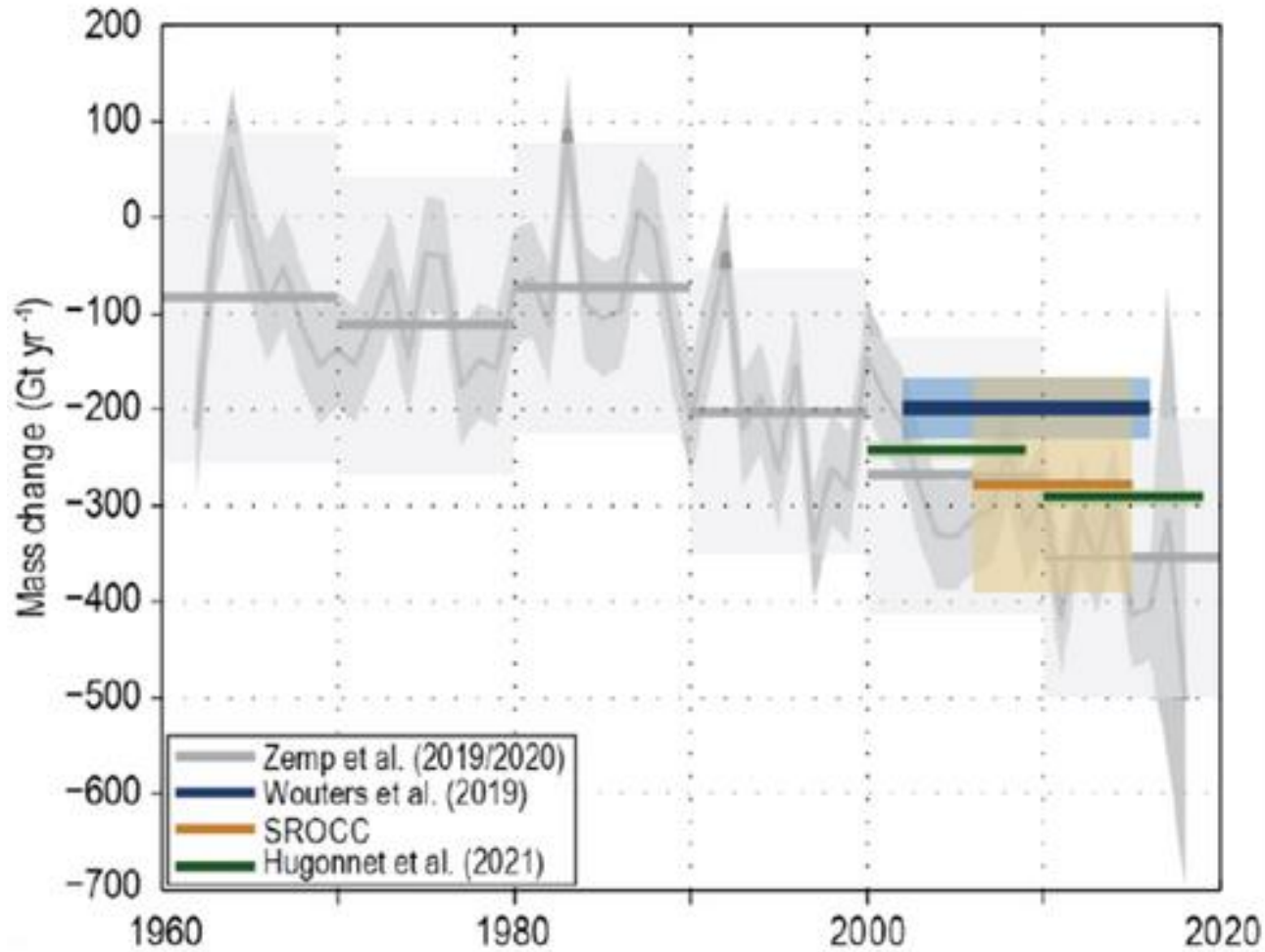


Ice Sheets

- The Greenland Ice Sheet and the Antarctic Ice Sheet have lost 4,890 Gt (billion tonnes) and 2,670 Gt of mass respectively over the period 1992–2020
- The rate of ice sheet loss increased by a factor of four between 1992-1999 and 2010-2019

OBSERVED CHANGES

Ice and Snow

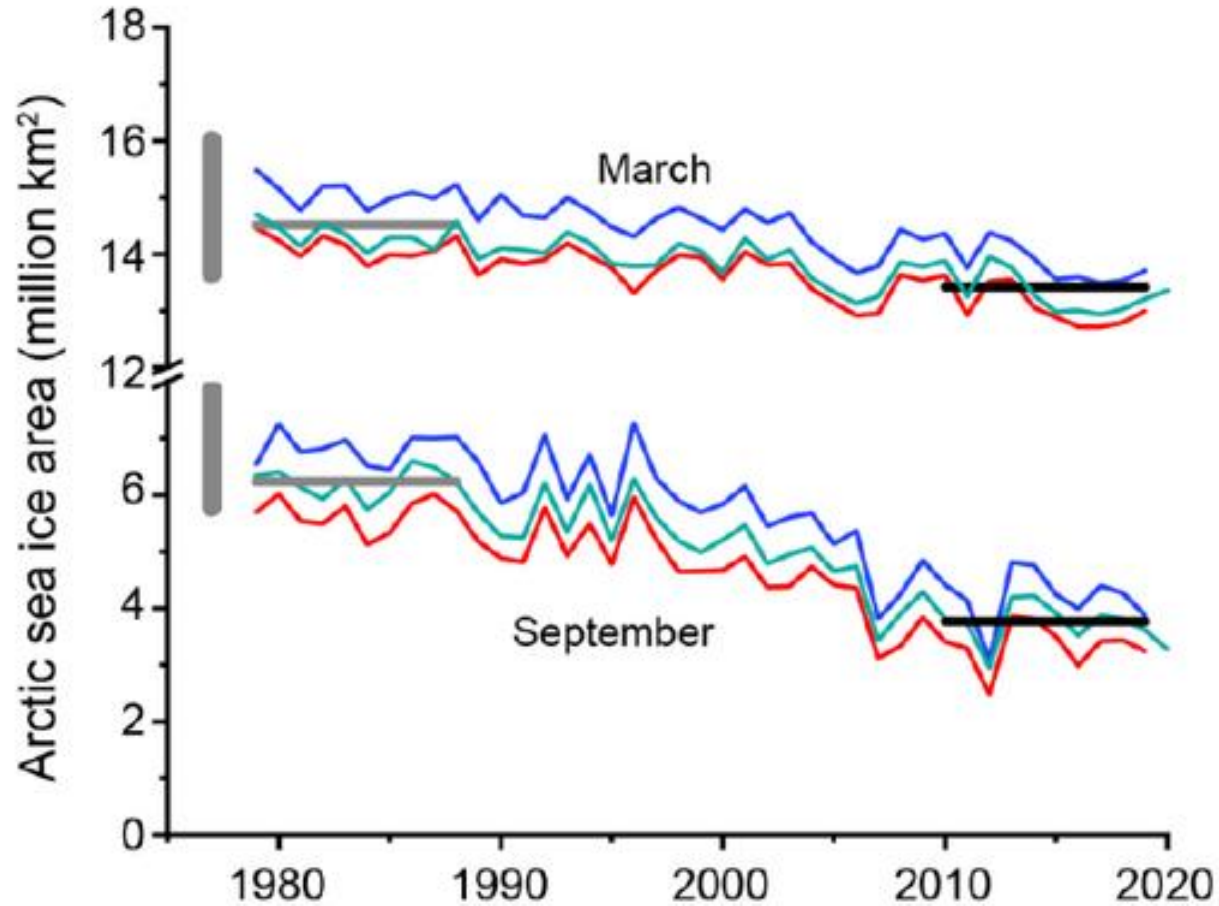


Glaciers

- Glaciers lost 6,200 Gt of mass over the period 1993-2019
- Global glacier retreat since the 1950s is unprecedented in at least the last 2,000 years

OBSERVED CHANGES

Ice and Snow

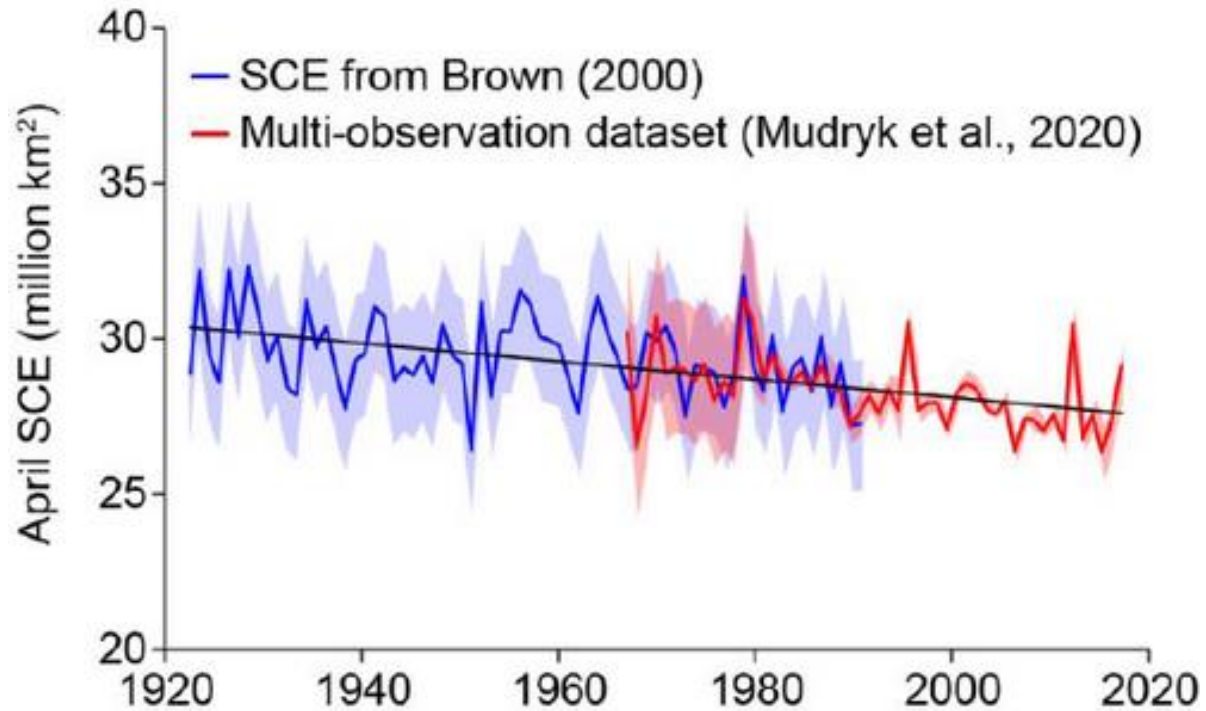


Arctic Sea Ice

- In 2011-2020, annual average Arctic sea ice area reached its lowest level since at least 1850
- Late summer Arctic sea ice area was smaller than at any time in at least the past 1,000 years

OBSERVED CHANGES

Ice and Snow



Snow Cover

- Northern Hemisphere spring snow cover extent has been decreasing since 1978, and there is high confidence that this trend extends back to 1950

PROJECTED CHANGES

What Does The Future Look Like?

CLIMATE PREDICTION vs PROJECTIONS

- Climate predictions and climate projections are both used to understand future climate conditions, but they differ in scope and purpose

Climate predictions

- Short-term forecasts (months to a few years) of specific climate phenomena, like seasonal temperature anomalies or precipitation patterns, based on current climate conditions and natural variability

Climate projections

- Long-term simulations (decades to centuries) of how the climate might change under different scenarios of greenhouse gas emissions and socioeconomic development factors, exploring potential future climates rather than predicting specific events
 - **Scenarios** describe potential future trajectories of greenhouse gas emissions and societal development

CLIMATE PREDICTION vs PROJECTIONS

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Climate predictions

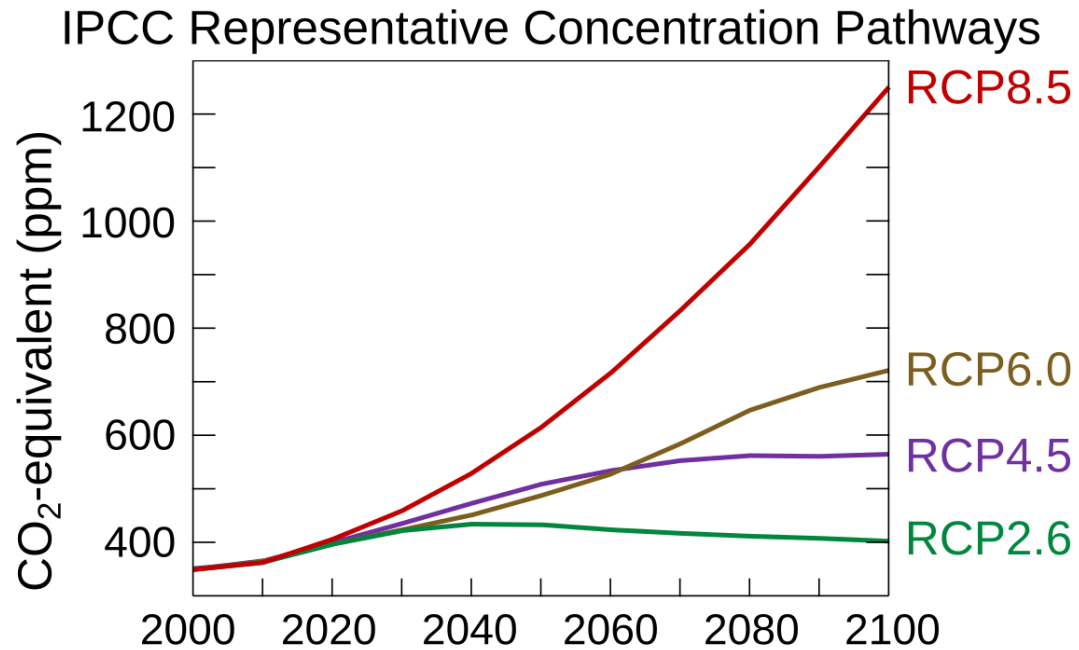
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PROJECTED CHANGES

Scenarios – Potential Futures

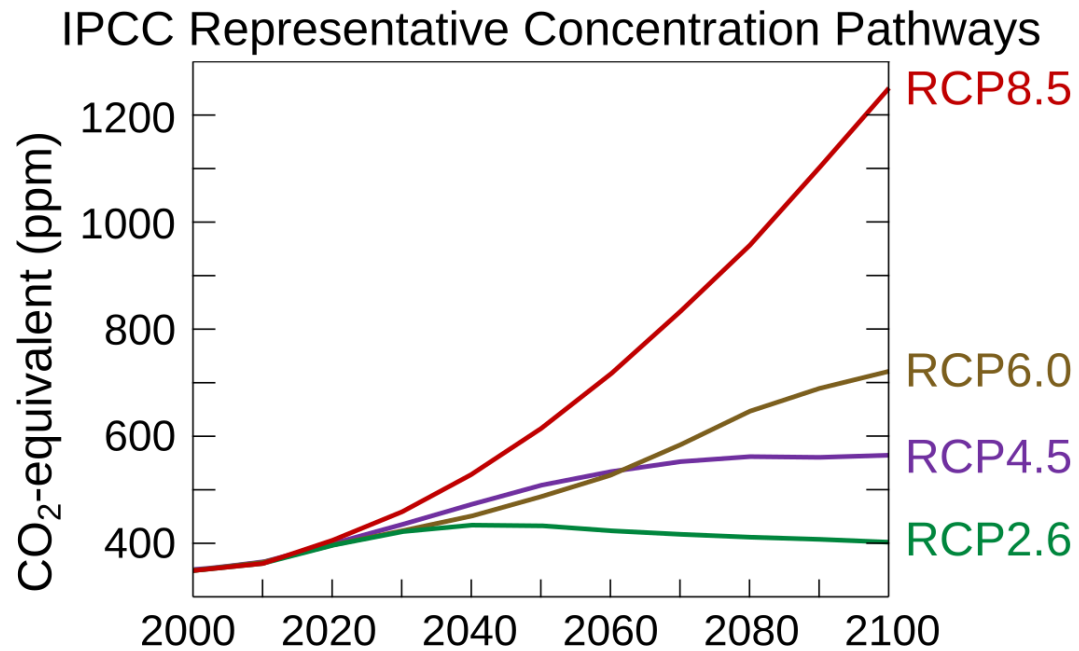


Representative Concentration Pathways (RCPs)

- Describe different trajectories of greenhouse gas concentrations in the atmosphere
- Used as inputs for Global Climate Models (GCMs) to simulate how the climate will respond to various levels of radiative forcing
- Labeled based on the level of radiative forcing that they represent by the end of the 21st century
- For example, RCP2.6 represents a scenario with relatively low radiative forcing, while RCP8.5 represents a scenario with high radiative forcing

PROJECTED CHANGES

Scenarios – Potential Futures



- **RCP2.6:** A future with very low GHG emissions
- **RCP4.5:** Intermediate stabilization of GHG concentrations by mid-century
- **RCP6.0:** Higher intermediate stabilization of GHGs concentrations towards the end of the century
- **RCP8.5:** High emissions (business as usual), where GHGs continue to increase

PROJECTED CHANGES

Scenarios – Potential Futures



Shared Socioeconomic Pathways (SSPs)

- Describe different potential futures for socioeconomic development, including factors like population growth, economic development, and energy use

PROJECTED CHANGES

Scenarios – Potential Futures

- There are five SSP scenarios:



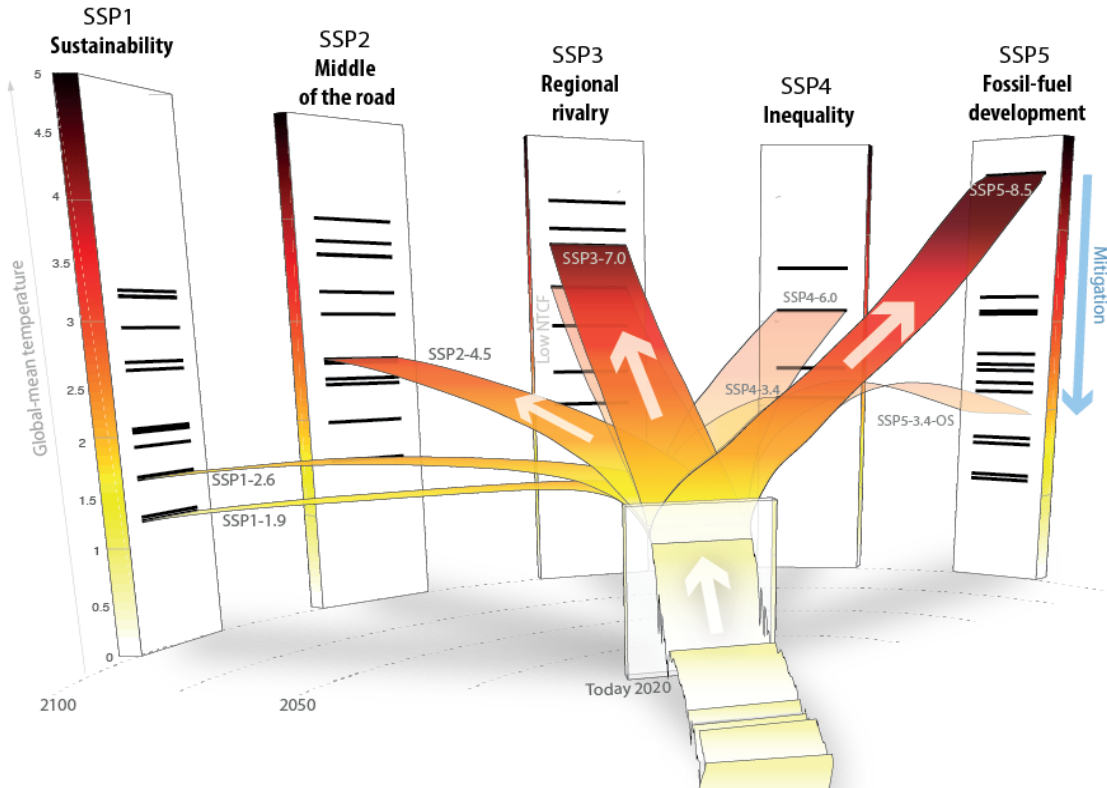
- **SSP1 – Sustainability:** towards a more sustainable path, emphasizing inclusive development and environmental awareness
- **SSP2 – Middle of the Road:** continuation of current trends with moderate challenges to mitigation and adaptation
- **SSP3 – Regional Rivalry:** Increased nationalism and conflicts, leading to slow economic growth and high challenges to both mitigation and adaptation
- **SSP4 – Inequality:** Increasing inequality, within and between countries, with high challenges to mitigation and adaptation
- **SSP5 – Fossil-fueled Development:** Rapid economic growth and fossil fuel consumption, leading to high emissions and challenges to mitigation

PROJECTED CHANGES

Scenarios – Potential Futures

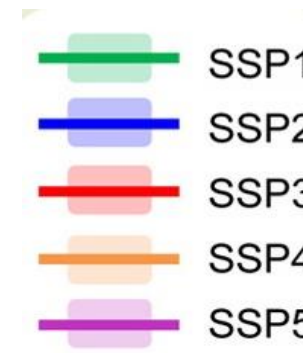
Combined SSP – RCP Scenarios

- Combined to provide a more comprehensive picture of potential future climate change impacts, as socioeconomic factors significantly influence greenhouse gas emissions
- The SSPs are designed to complement the RCPs by exploring a wider range of potential futures, including those with varying levels of mitigation efforts and adaptation challenges

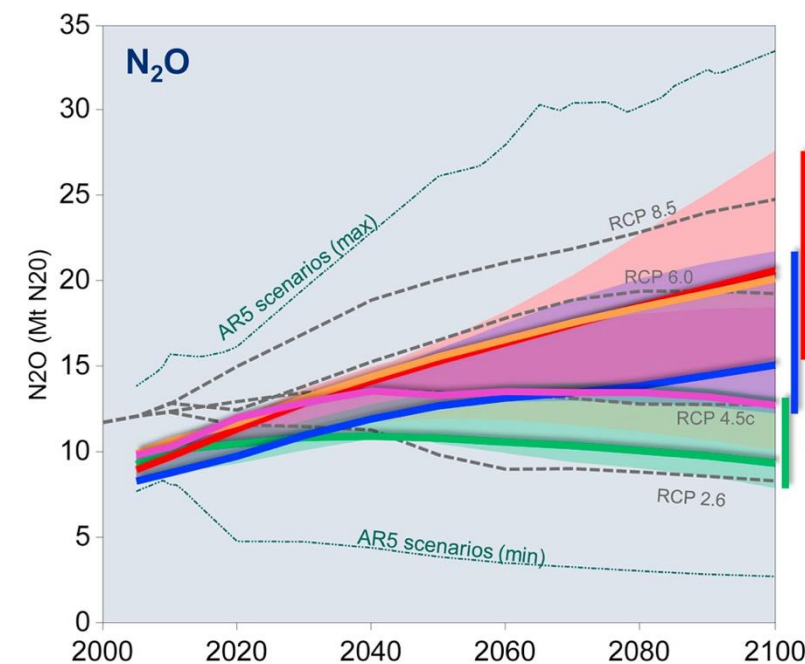
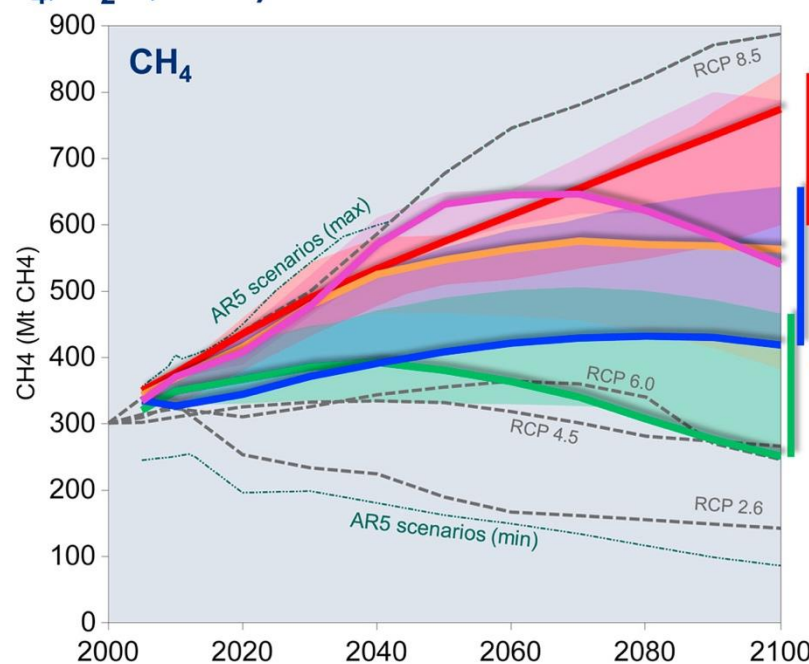
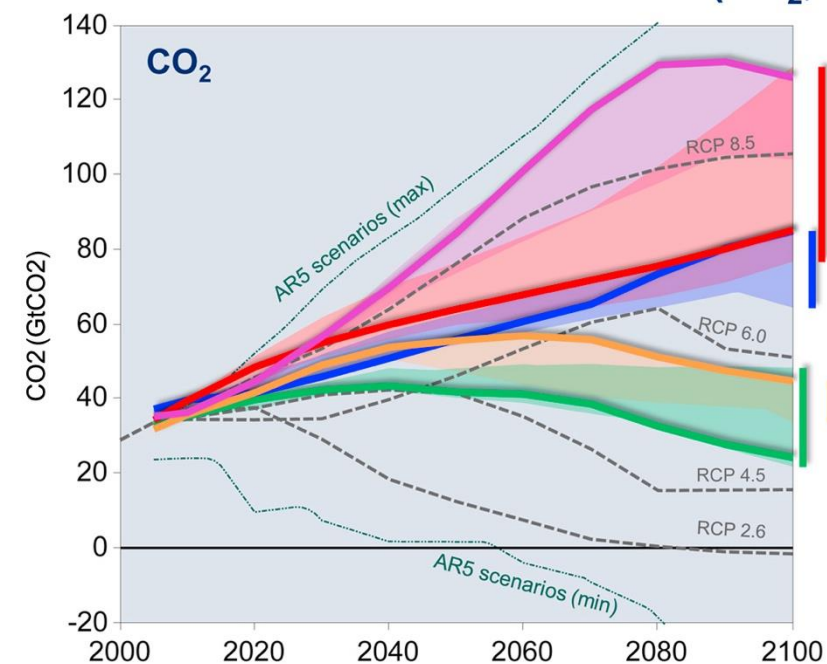


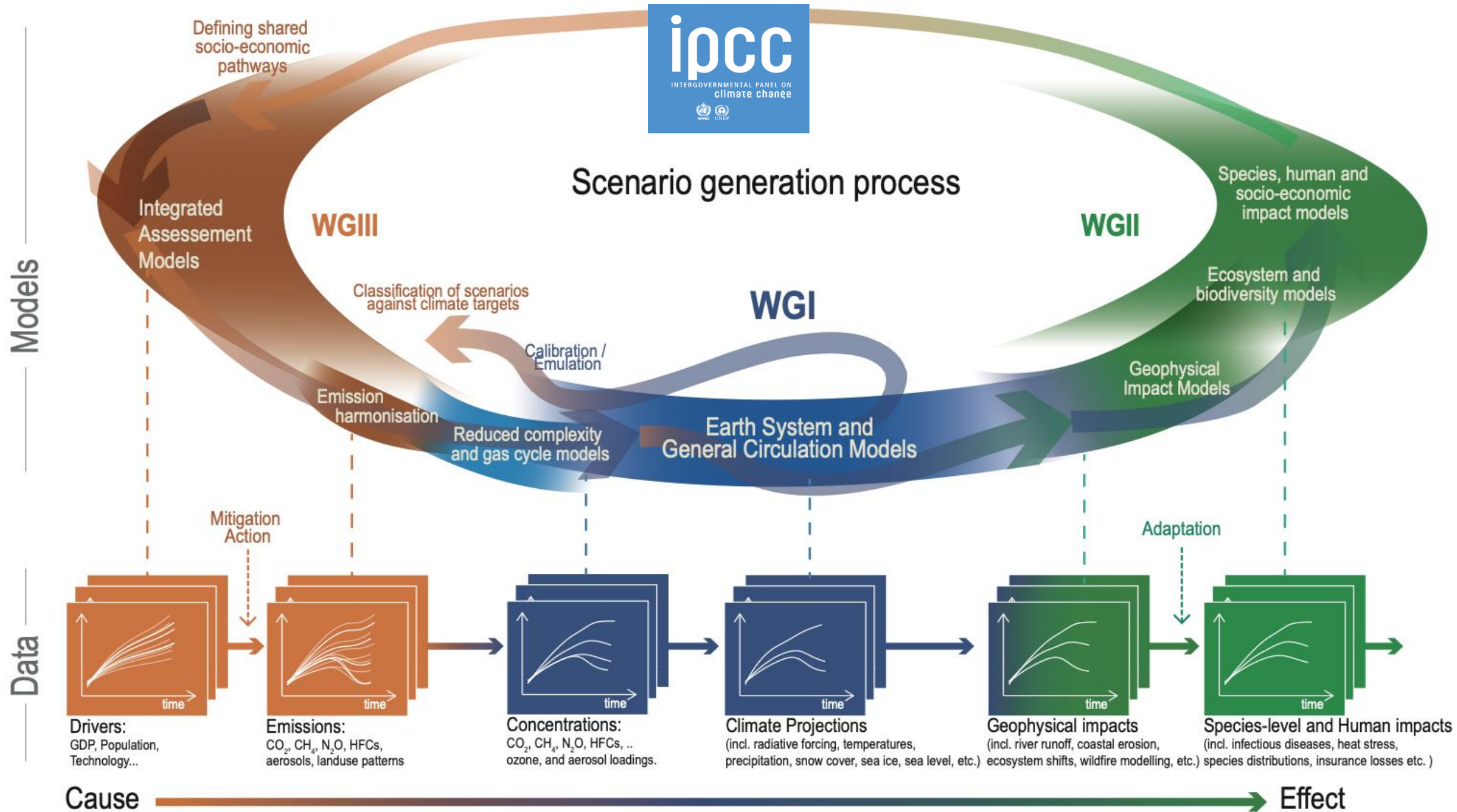
PROJECTED CHANGES

Scenarios – Potential Futures



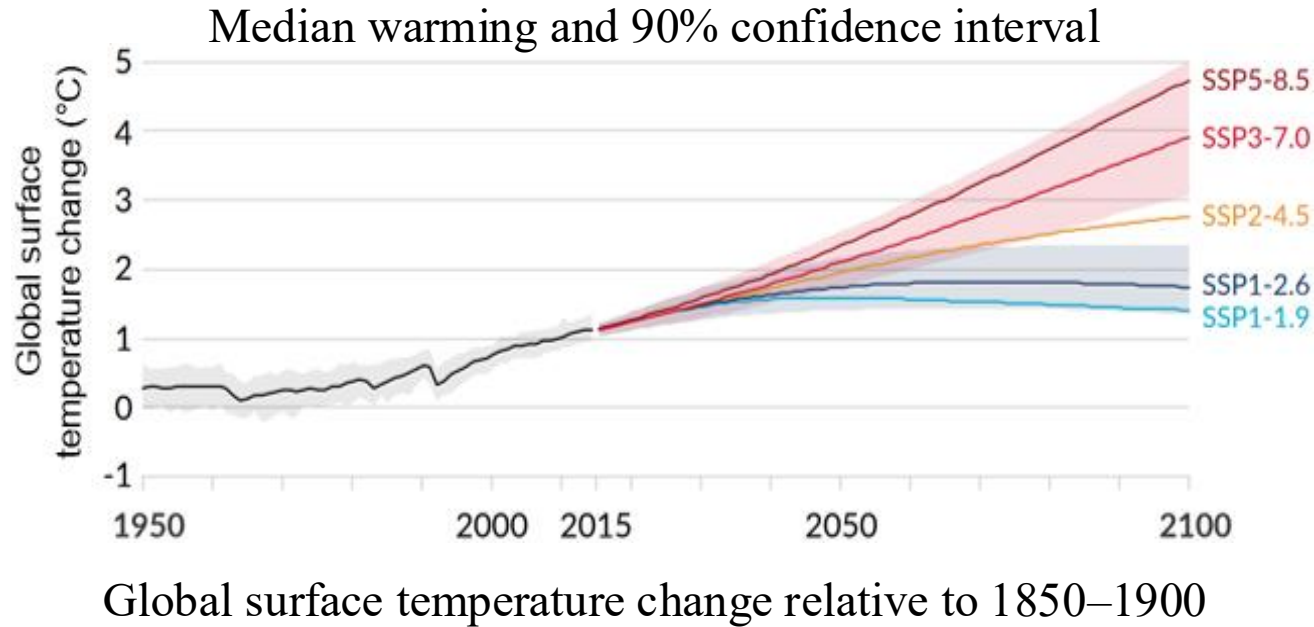
Greenhouse Gas Emissions (CO₂, CH₄, N₂O, etc..)





PROJECTED CHANGES

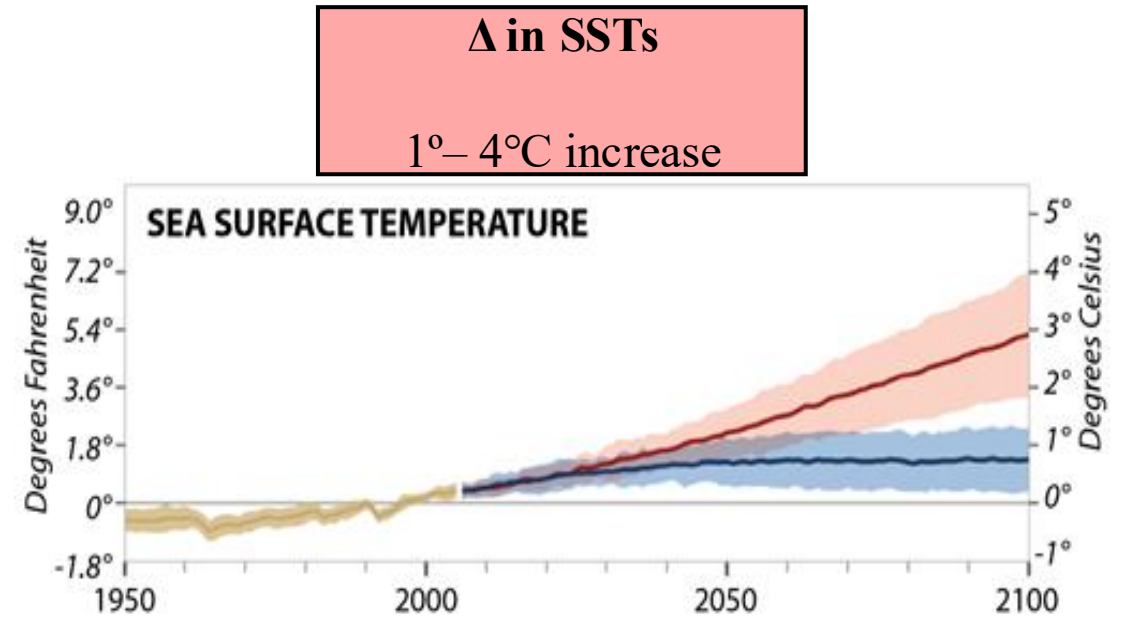
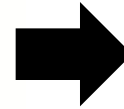
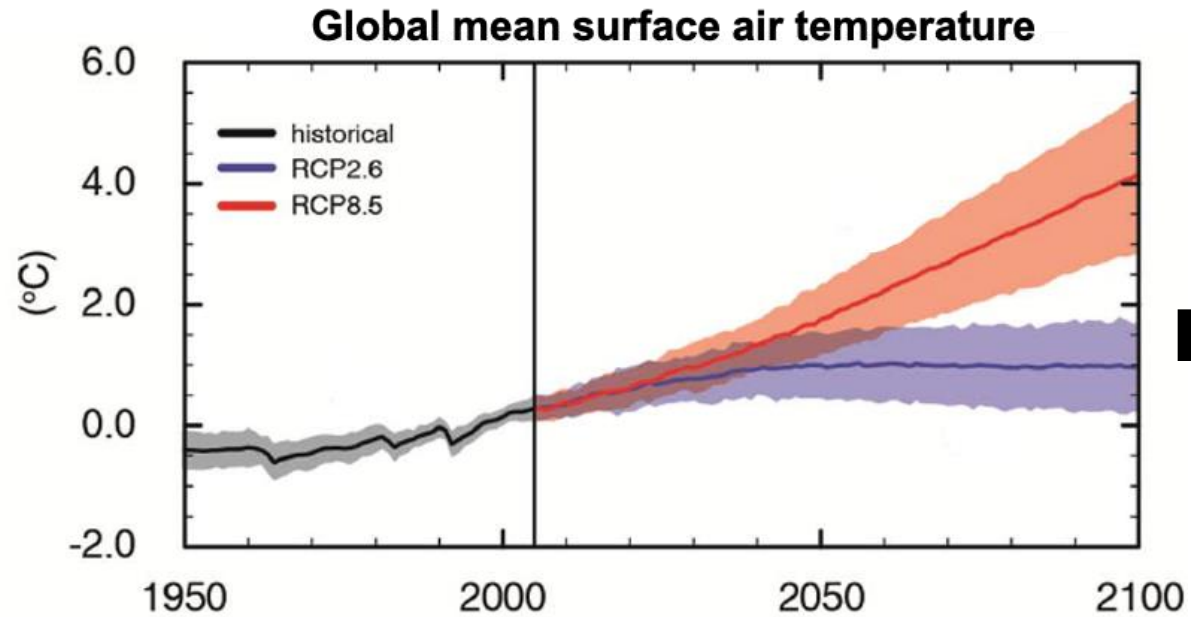
Changes in Temperature



- Global surface temperature will continue to increase until at least the mid-century under all emissions scenarios
 - Global warming of 1.5°C and 2°C will be exceeded during the 21st century unless deep reductions in CO₂ and other GHG emissions occur in the coming decades
- Compared to 1850–1900, the global surface temperature averaged over 2081–2100 is very likely to be higher by:
 - 1.0-1.8°C (SSP1-1.9)
 - 2.1-3.5°C (SSP2-4.5)
 - 3.3-5.7°C (SSP5-8.5)

PROJECTED CHANGES

Changes in Temperature



Δ in SSTs

1°– 4°C increase

PROJECTED CHANGES

Changes in Temperature

Regional Nearshore Cooling

South African west coast:

- 0.5°C cooling per decade
- Could end up with a 4°C cooling by 2100

Coastal oceanic climate change and variability from 1982 to 2009 around South Africa

M Rouault^{1*}, B Pohl² and P Penven³

¹ Department of Oceanography, Marine Research Institute, University of Cape Town, Rondebosch 7701, South Africa

² Centre de Recherches de Climatologie, CNRS / Université de Bourgogne, 6 boulevard Gabriel, 21000 Dijon, France

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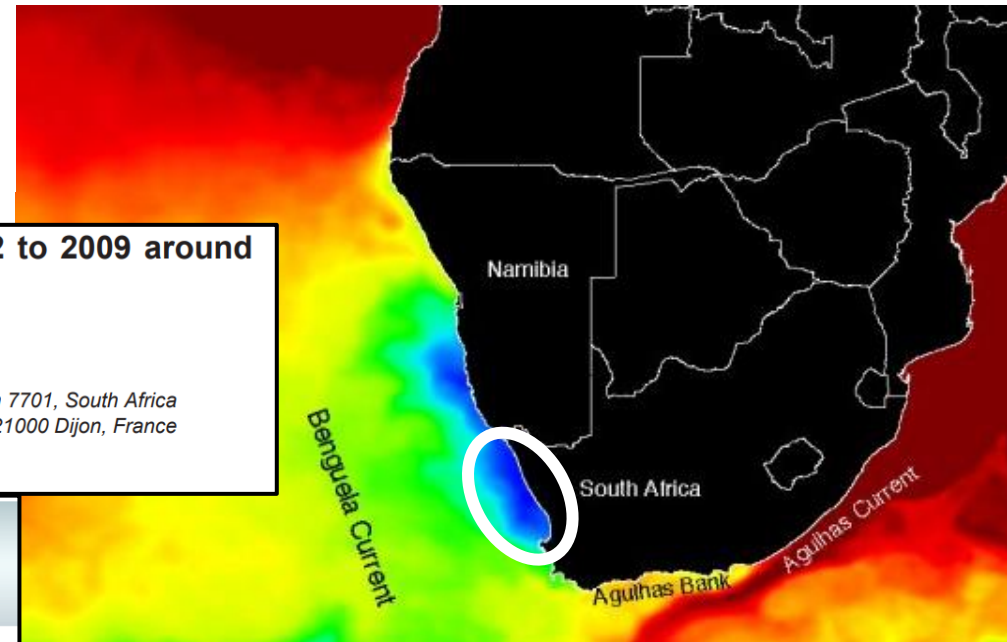
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Three decades of high-resolution coastal sea surface temperatures reveal more than warming

Fernando P. Lima^{1,2} & David S. Wetthey¹

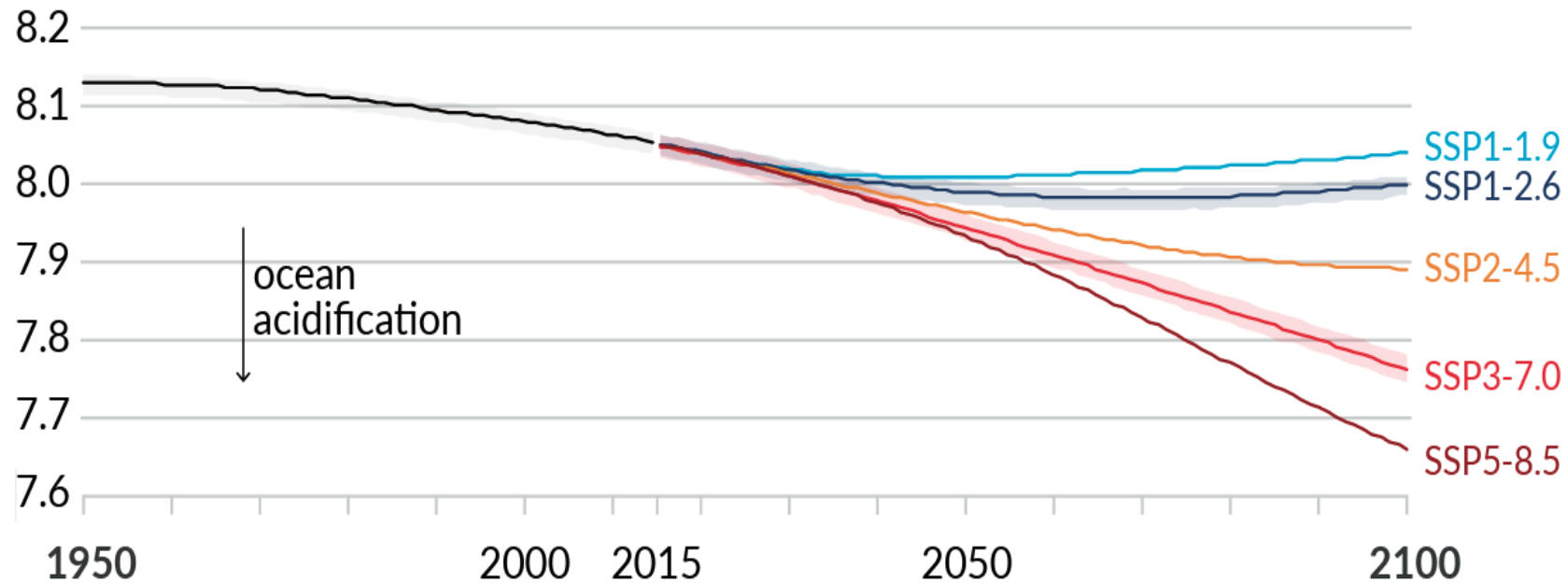


Upwelling indices for comparative ecosystem studies: Variability in the Benguela Upwelling System

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PROJECTED CHANGES

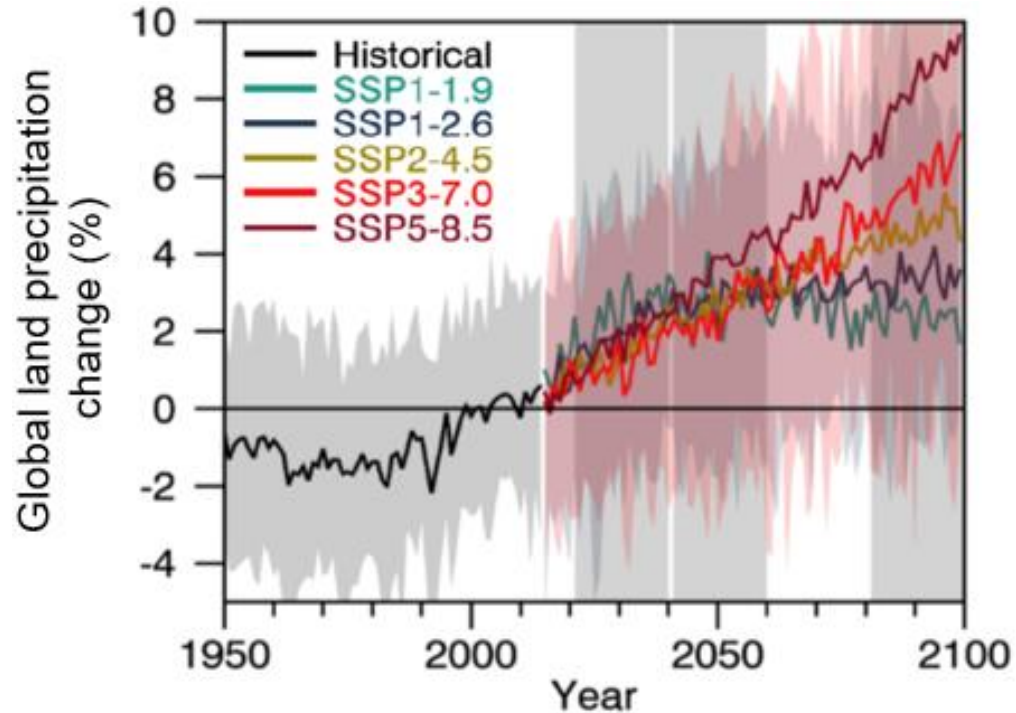
Ocean Acidification



- Model mean and model spread in the decrease in surface ocean pH from 1986–2005 to 2081–2100:
 - 0.065 (0.06–0.07) for RCP2.6
 - 0.145 (0.14–0.15) for RCP4.5
 - 0.203 (0.20–0.21) for RCP6.0
 - 0.31 (0.30–0.32) for RCP8.5

PROJECTED CHANGES

Changes in Precipitation



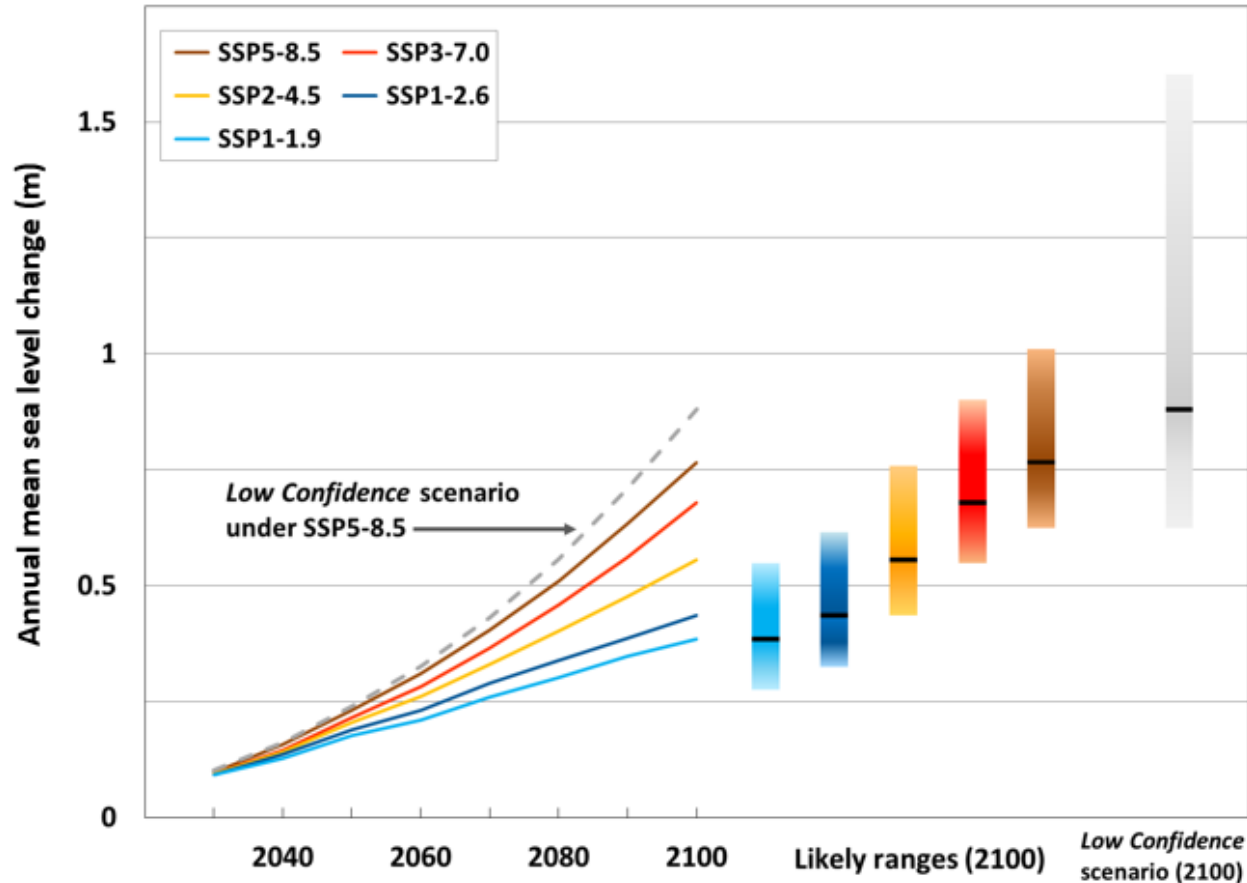
- Projected to become more variable over most land regions within seasons and from year to year due to continued global warming
- It is very likely that heavy precipitation events will intensify and become more frequent in most regions with additional global warming
- At the global scale, extreme daily precipitation events are projected to intensify by about 7% for each 1°C of global warming

Global land precipitation change relative to 1995–2014

- Compared to 1995–2014, the average annual global land precipitation in 2081–2100 is projected to increase by:
 - 0-5% (SSP1-1.9)
 - 1.5-8% (SSP2-4.5)
 - 1-13% (SSP5-8.5)

PROJECTED CHANGES

Changes in Mean Sea Level

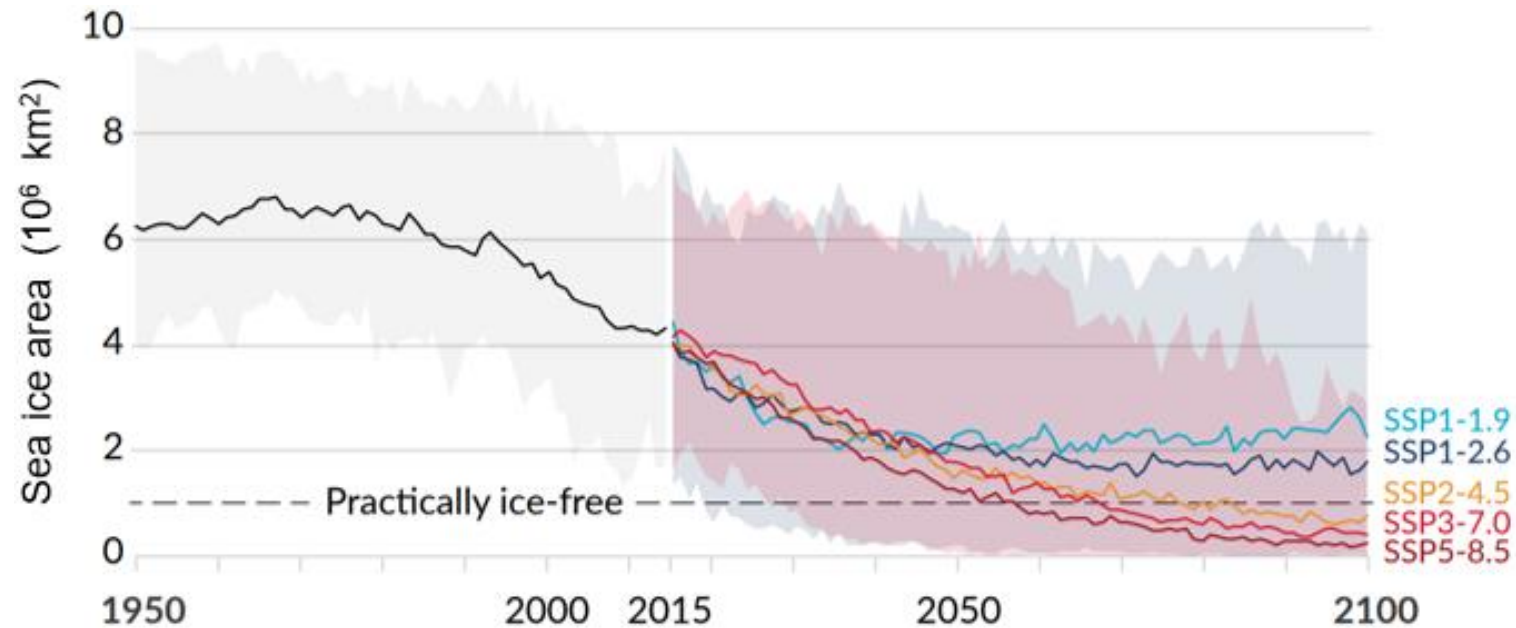


Global mean sea level change relative to 1995–2014

- Global mean sea level will continue to rise over the 21st century
- Relative to 1995–2014, the likely global mean sea level rise by 2100 is:
 - 0.28-0.55 m (SSP1-1.9)
 - 0.44-0.76 m (SSP2-4.5)
 - 0.63-1.01 m (SSP5-8.5)
- *Low Confidence* scenario under SSP5-8.5 – impact of uncertain ice sheet processes

PROJECTED CHANGES

Changes in Ice and Snow



- The Arctic is likely to be sea ice free in September at least once before 2050
- Mountain and polar glaciers are committed to continue melting for decades or centuries
- Continued ice loss over the 21st century for the Greenland Ice Sheet and likely for the Antarctic Ice Sheet
- Additional warming is projected to further amplify loss of seasonal snow cover



THANK YOU